

While the pipeline produced meaningful improvements across all runs, several limitations were observed. Training results exhibited inconsistency across runs, likely due to the subtle and overlapping visual features that distinguish facial emotion classes — a challenge that may require careful, manual curation of the dataset to achieve more reliable performance. Additionally, because the training data is sourced externally, it may contain malicious or low-quality content; to mitigate this, the pipeline includes a filtering step to remove such samples before training. Finally, training time scales linearly with the number of target classes, which may become a practical bottleneck as the classification task is expanded beyond the three emotions evaluated here.

Conclusion & Future Scope

This project demonstrates that combining CLIP-based filtering with LLM orchestration creates a viable, fully automated pipeline for zero-shot data collection and model fine-tuning. Across all eight experimental runs, fine-tuning consistently delivered accuracy gains of +25.6 to +42.3 percentage points without any manually labeled data, with MobileNet-V3 achieving the best fine-tuned accuracy at 71.43%. Despite these results, training outcomes were inconsistent across runs due to the subtle visual overlap between emotion classes, external data sources introduced quality risks mitigated only partially by filtering, and training time scales linearly with the number of classes — all challenges that limit the pipeline's robustness at scale. Future work will focus on three key extensions: supporting user-uploaded images for hybrid datasets, integrating GPT-4V for automated label verification and active learning, and expanding beyond image classification to object detection frameworks such as YOLO — moving the system from a research prototype toward a general-purpose tool for prompt-driven model development.